

```

ubuntu@test-vm1:~$ sudo apt-get update
Err:1 http://archive.ubuntu.com/ubuntu bionic InRelease
  407 Proxy Authentication Required [IP: 10.201.51.54 8080]
Err:2 http://archive.ubuntu.com/ubuntu bionic-security InRelease
  407 Proxy Authentication Required [IP: 10.201.51.54 8080]
Err:3 https://download.docker.com/linux/ubuntu bionic InRelease
  Reading from proxy failed - read (115: Operation now in progress) [IP: 10.201.51.54 8080]
Err:4 http://archive.ubuntu.com/ubuntu bionic-updates InRelease
  407 Proxy Authentication Required [IP: 10.201.51.54 8080]
Reading package lists... Done
N: See apt-secure(8) manpage for repository creation and user configuration details.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
E: The repository 'http://archive.ubuntu.com/ubuntu bionic InRelease' is no longer signed.
E: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/bionic/InRelease 407 Proxy Authentication Required [IP: 10.201.51.54 8080]
E: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/bionic-security/InRelease 407 Proxy Authentication Required [IP: 10.201.51.54 8080]
E: The repository 'http://archive.ubuntu.com/ubuntu bionic-security InRelease' is no longer signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
E: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/bionic-updates/InRelease 407 Proxy Authentication Required [IP: 10.201.51.54 8080]
E: The repository 'http://archive.ubuntu.com/ubuntu bionic-updates InRelease' is no longer signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
ubuntu@test-vm1:~$

```

Figure 1: Error showing proxy authentication is required

The example of the error is shown in Figure 1.

To solve this, ensure that the proxy has been set up properly. To make *apt-get* to work behind the proxy, we need to follow the two-step process mentioned below.

- a. Create a */etc/apt/apt.conf* file and add the following entries to the file:

```

$ vi /etc/apt/apt.conf
Acquire::http::Proxy "http://username:password@proxy:port/";
Acquire::https::Proxy "https://username:password@
proxy:port/";

```

- b. Create a */etc/apt/apt.conf.d/proxy.conf* file and add the following entries in the file:

```

$ vi /etc/apt/apt.conf.d/proxy.conf
Acquire::http::Proxy "http://username:password@proxy:port/";
Acquire::https::Proxy "https://username:password@
proxy:port/";

```

- 2) *Problem:* Sometimes one keeps on facing errors even after setting *apt config*, as mentioned earlier. The installation or update will fail and might not work again behind the proxy or sometimes, the pull operation of the Docker image will have issues.

The following step will resolve this issue.

Ensure that the proper DNS is set up by entering the name servers in the file */etc/resolv.conf*:

```

$ vi /etc/resolv.conf
nameserver xx.xx.xx.xx
nameserver xx.xx.xx.xx
nameserver 127.0.0.53
search company.com

```

 **Note:** This <IP address> is based on the host IP and gateway used, and will differ based on the configuration.

Pre-configured proxy on container images (migration/normal set-up scenarios)

When the container image is built and carried to a different environment, it should work fine (like DNS, routing, etc).

- 1) *Problem:* In some cases, *apt-get* or *internet* will not work during the building of container images.

This problem can be solved in the following way. The environment variable should be set up properly by using the following procedure.

- a. Verify that the *http_proxy*, *https_proxy*, *HTTP_PROXY* and *HTTPS_PROXY* variables are set up by using the following *echo* command:

```

ubuntu@test-vm1:/etc/apt$ echo $http_proxy
http://username:password@proxy:port/

```

Ensure that the output is as shown above. If the output is empty, then proceed to set up the environment variable as shown in Step 2 below.

- b.


```

# export http_proxy=http://username:password@proxy:port
# export https_proxy=https:// username:password@proxy:port

```

```

# Docker Upstart and SysVinit configuration file
#
# THIS FILE DOES NOT APPLY TO SYSTEMD
#
# Please see the documentation for "systemd drop-ins":
# https://docs.docker.com/engine/admin/systemd/
#
# Customize location of Docker binary (especially for development testing).
#DOCKER="/usr/local/bin/docker"
#
# See DOCKER_OPTS to modify the daemon startup options.
#DOCKER_OPTS="--dns 8.8.8.8 --dns 8.8.4.4"
DOCKER_OPTS="--dns 10.200.50.100 --dns 10.200.51.100"
#
# If you need Docker to use an HTTP proxy, it can also be specified here.
#export http_proxy="http://127.0.0.1:3123/"
http_proxy=http://username:password@proxy:port
https_proxy=https://username:password@proxy:port
HTTP_PROXY=http://username:password@proxy:port
HTTPS_PROXY=https://username:password@proxy:port
#
# This is also a handy place to tweak where Docker's temporary files go.
#export DOCKER_TEMPDIR="/mnt/highdrive/docker-temp"

```

Figure 2: Adding proxy details to the configuration file